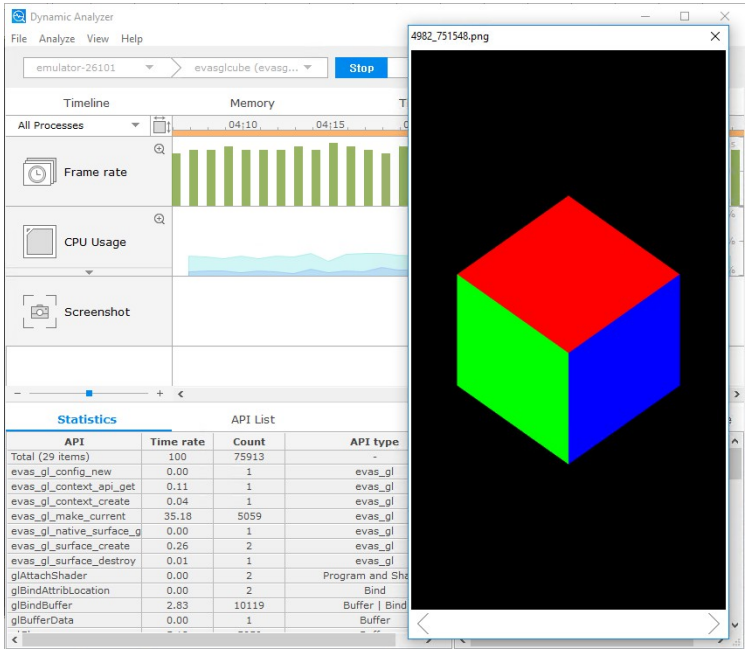
OpenGL

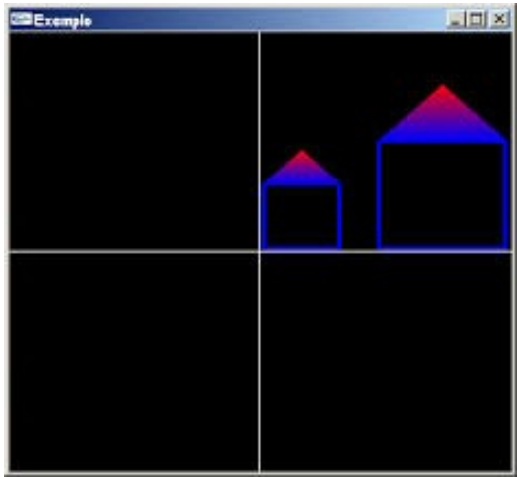
Tratamento de Eventos via Teclado

1.Introdução

Objetivo

- Criar um quadrado que se move com teclas W, A, S, D.
- Utilizar **eventos de teclado (teclas simples)** com **GLUT**.
- Entender a estrutura básica de um programa OpenGL com tratamento de eventos.

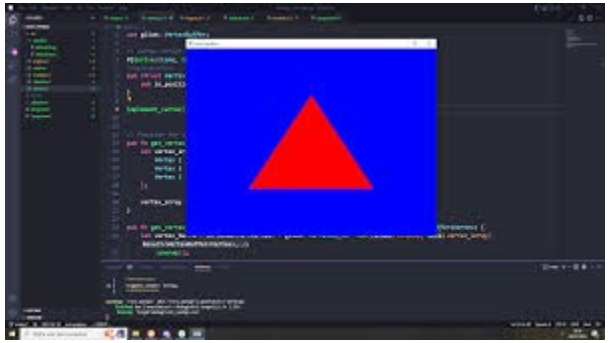


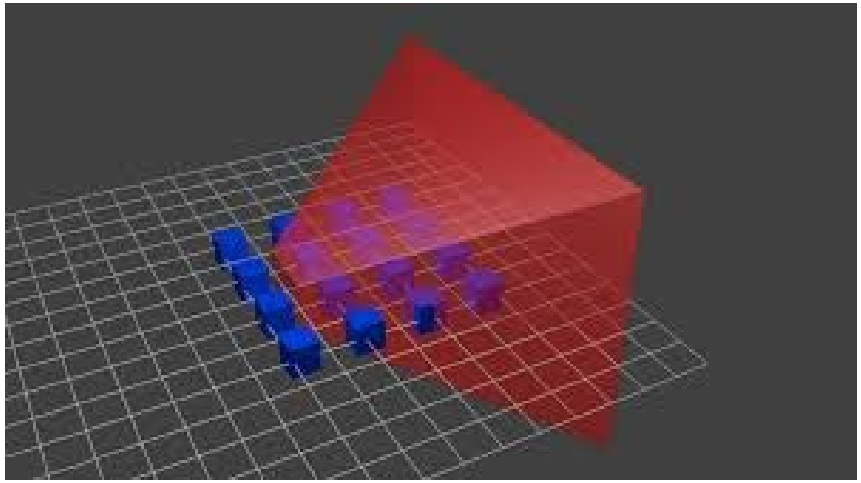


2. Estrutura do Código

Componentes principais

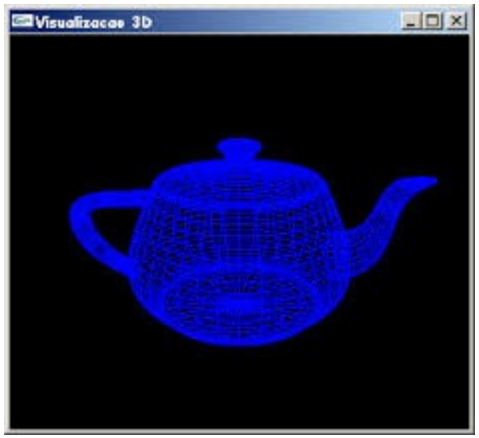
- display(): desenha a cena.
- keyboard(): trata a entrada do teclado.
- init(): configurações iniciais de OpenGL.
- main(): cria a janela e inicia o loop principal.



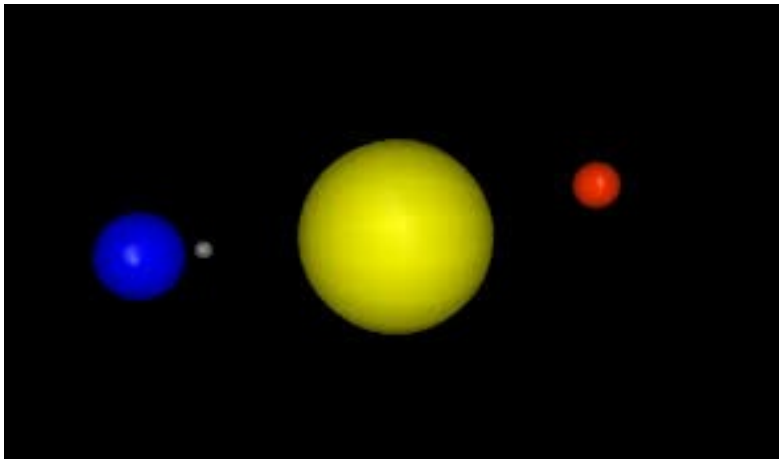


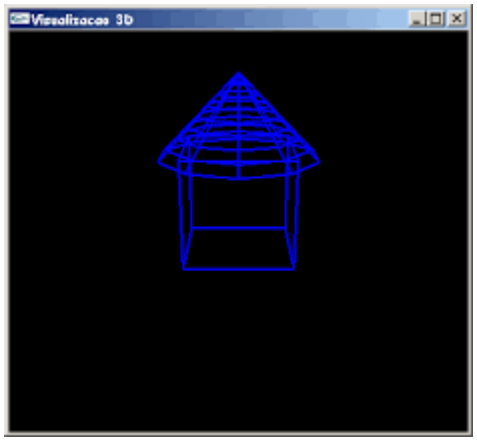
3. Variáveis Globais

```
float x = 0.0f;

float y = 0.0f;

float step = 0.1f;
```



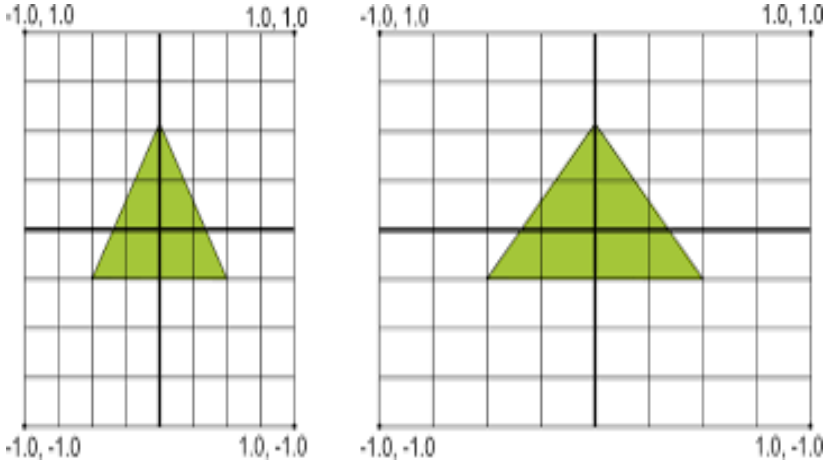
- **x** e **y**: posição do objeto.
- **step**: valor de deslocamento a cada tecla pressionada.





4. Função Display

```
void display() {
    glClear(GL_COLOR_BUFFER_BIT);
    glLoadIdentity();
    glTranslatef(x, y, 0.0f);

    glBegin(GL_QUADS);
        glColor3f(1.0f, 0.0f, 0.0f);
        glVertex2f(-0.1f, -0.1f);
        glVertex2f( 0.1f, -0.1f);
        glVertex2f( 0.1f,  0.1f);
        glVertex2f(-0.1f,  0.1f);
    glEnd();

    glutSwapBuffers();
}
```

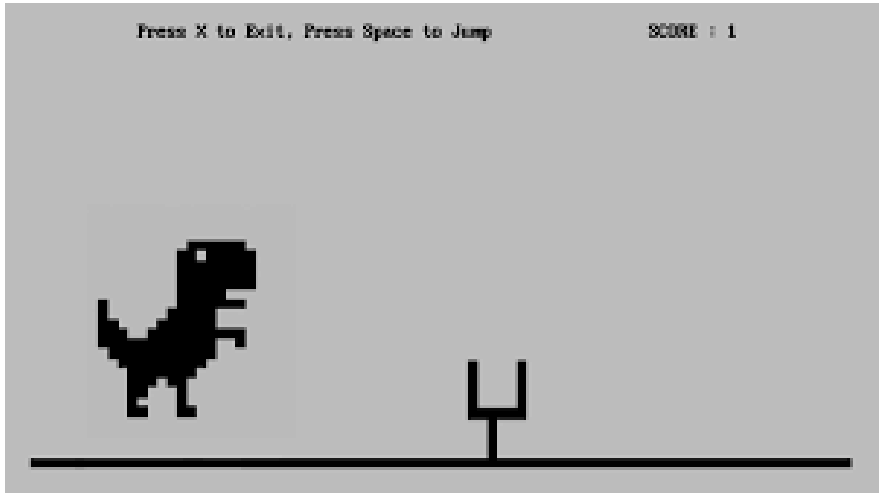


- **Limpa a tela.**
- **Aplica a translação (posição).**
- **Desenha o quadrado vermelho.**
- **Atualiza a janela.**

5. Função keyboard()

```
void keyboard(unsigned char key, int xMouse, int yMouse) {
    switch (key) {
        case 'w': y += step; break;
        case 's': y -= step; break;
        case 'a': x -= step; break;
        case 'd': x += step; break;
        case 27: exit(0); // ESC
    }
    glutPostRedisplay();
}
```

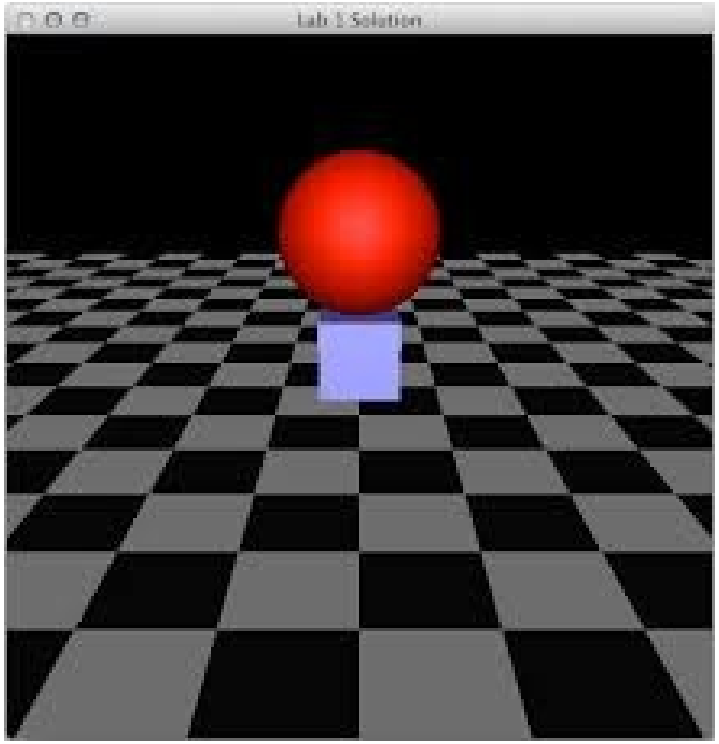
- **Detecta a tecla pressionada.**
- **Move o quadrado mudando x ou y.**
- **ESC (27) encerra o programa.**



6. Função init()

```
void init() {
    glClearColor(0.0, 0.0, 0.0, 1.0);
    glMatrixMode(GL_PROJECTION);
    glLoadIdentity();
    gluOrtho2D(-1.0, 1.0, -1.0, 1.0);
}
```

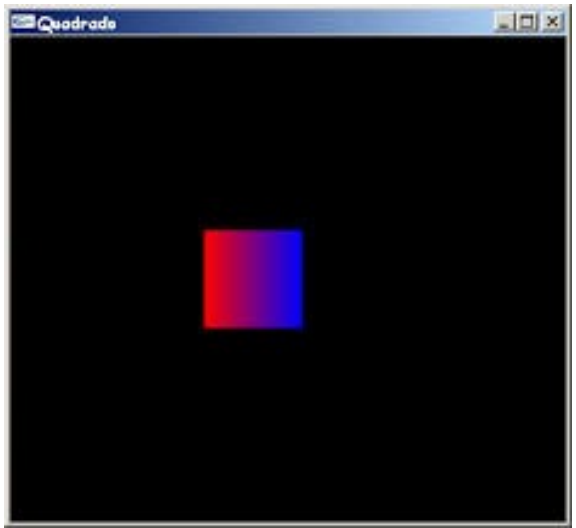
- **Define cor de fundo preta**
- **Carrega matriz identidade**
- **Configura a projeção ortográfica (2D).**



7. Função principal

```
int main(int argc, char** argv) {
    glutInit(&argc, argv);
    glutInitDisplayMode(GLUT_DOUBLE | GLUT_RGB);
    glutInitWindowSize(500, 500);
    glutCreateWindow("Controle com Teclado - OpenGL");

    init();
    glutDisplayFunc(display);
    glutKeyboardFunc(keyboard);

    glutMainLoop();
}
```

- **Inicializa GLUT.**
- **Cria a janela de 500x500 pixels.**
- **Registra funções de desenho e teclado.**
- **Inicia o loop principal.**

7. Código inteiro

```
#include <GL/glut.h>

// Posição do quadrado
float x = 0.0f;
float y = 0.0f;
float step = 0.1f;

// Função para desenhar a cena
void display() {
    glClear(GL_COLOR_BUFFER_BIT);
    glLoadIdentity();

    glTranslatef(x, y, 0.0f);
    glBegin(GL_QUADS);
        glColor3f(1.0f, 0.0f, 0.0f);
        glVertex2f(-0.1f, -0.1f);
        glVertex2f( 0.1f, -0.1f);
        glVertex2f( 0.1f,  0.1f);
        glVertex2f(-0.1f,  0.1f);
    glEnd();

    glutSwapBuffers();
}

void keyboard(unsigned char key,
              int xMouse, int yMouse) {
    switch (key) {
        case 'w': y += step; break;
        case 's': y -= step; break;
        case 'a': x -= step; break;
        case 'd': x += step; break;
        case 27: exit(0);   // tecla ESC
    }
    glutPostRedisplay(); // Solicita redesenho
}

void init() {
    glClearColor(0.0, 0.0, 0.0, 1.0);
    glMatrixMode(GL_PROJECTION);
    glLoadIdentity();
    gluOrtho2D(-1.0, 1.0, -1.0, 1.0);

}

// Função principal
int main(int argc, char** argv) {
    glutInit(&argc, argv);
    glutInitDisplayMode(GLUT_DOUBLE | GLUT_RGB);
    glutInitWindowSize(500, 500);
    glutCreateWindow("Controle com Teclado - OpenGL");

    init();
    glutDisplayFunc(display);
    glutKeyboardFunc(keyboard);
    glutMainLoop();
    return 0;
}
```